Xiaoqi Zhao 赵骁骐

NTU AI4X Principal Investigator

NRF Postdoctoral Award · NTU AI-for-X Postdoctoral Fellowship

Founding Partner · X3000 INSPECTION (Startup, Series B Funded)

Contact

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Google Scholar

github.com/Xiaoqi-Zhao-DLUT

Last updated: October 2026

My long-term research vision is to build a general-purpose visual intelligence engine under the AI-for-X paradigm. This vision is structured around three interconnected layers: **Pattern, Unification, AI4X Applications.** At the pattern level, I investigate self-driven learning mechanisms such as self-supervised, open-vocabulary, prompt-based, and in-context learning to establish scalable and transferable learning principles that reduce supervision dependency and enhance generalization. At the unification level, I study how diverse visual tasks, concepts, and modalities can be integrated within shared architectures and representations, enabling unified multi-task modeling, context-dependent concept understanding, and image-video multi-modal perception under a coherent semantic framework. Built upon these foundations, the AI4X application layer extends the engine into real-world domains such as healthcare and industry, validating its robustness in complex, domain-specific environments while **continuously expanding the scope of X**. Ultimately, progressing toward visual AGI requires iterative strengthening of learning patterns, deeper unification across tasks and modalities, and sustained expansion of AI4X domains to achieve multi-source input processing, multi-task learning, multi-modal fusion, cross-domain generalization, and a balance between expert-level performance and unified modeling.

Education, Entrepreneurship & Professional Experience

Nanyang Technological University — Singapore

AI4X Principal Investigator (Mar. 2026 –)
NRF Postdoctoral Award (PI, SGD $250,000) · NTU AI-for-X Postdoctoral Fellowship
Collaboration with Prof. Weisi Lin (IEEE Fellow) & Prof. Dacheng Tao (ACM & IEEE Fellow)

Yale University — New Haven, USA

Postdoctoral Researcher (Feb. 2025 – Mar. 2026)
Collaboration with Prof. Xiaofeng Liu & Prof. Georges El Fakhri (AIMBE & IEEE Fellow)

X3000 INSPECTION (Startup, Series B Funded) — Beijing, China

Founding Partner & Equity Stake Holder (Jan. 2025 –)

X3000 INSPECTION (Startup, Series B Funded) — Beijing, China

Founding Partner & CTO & Chief AI Scientist (Sep. 2021 – Oct. 2024)
Led X3000Former / X3000NeXt foundational models; serving CATL, Geely, Chery, CALB, EVE Energy

Dalian University of Technology (Project 985) — Liaoning, China

Ph.D. Signal and Information Processing, IIAU-Lab (Sep. 2019 – Oct. 2024)
Advisors: Prof. Lihe Zhang & Prof. Huchuan Lu (IEEE Fellow)

Dalian University of Technology (Project 985) — Liaoning, China

B.E. Electronic Information Engineering (Sep. 2015 – Jun. 2019)

Research Interests

[01] Unified Visual Perception and Reasoning

- Segment Any Concept
- Context-Dependent Multi-Concept Understanding
- Unified Multi-modal / Image-Video / Multi-view Models

[02] Self-driven Learning

- Self-supervised / Unsupervised Learning
- Few-shot / In-context / Prompt Learning
- Open-Vocabulary Learning
- Foundation Model Evaluation & Adaptation

[03] AI4Industry

- Battery Management System
- Power Battery Detection & Generation with Ultra-dense Plates (600+ plates)
- High-speed Industrial CT Reconstruction
- Unsupervised Anomaly Detection
- Industrial Blind Image Enhancement

[04] AI4Health

- Medical Diagnosis with Incomplete Modalities
- PET-CT-MRI Fusion
- Medical Generalist Models

Honors and Awards

2026 **Stanford–Elsevier World's Top 2% Scientists**

2026 **MICCAI Young Scientist Publication Impact Award Finalist** (MICCAI'21-25 five-year publication pool; 9 / 11,939 ≈ 0.08%)

2026 **Singapore NRF Postdoctoral Award (PI)** — Top-tier Singapore Government Fellowship; SGD $250,000; up to 4-year support toward Assistant Professorship

2026 **NTU AI-for-X Postdoctoral Fellowship**

2025 **World Artificial Intelligence Conference (WAIC) Yunfan Award — Rising Star Nomination** (10 winners worldwide)

2025 Profiled by ***People's Daily*** **《人民日报》** as Outstanding PhD Student Representative (100 nationwide; 1 from DUT)

2021–25 **AI 2000 Most Influential Scholars** by AMiner in Computer Vision (5 consecutive years)

2022, 24 **National PhD Scholarship** (×2), Ministry of Education in China

2023 **Gold Award** · 8th China International "Internet+" Innovation & Entrepreneurship Competition, Company/Project: X3000 INSPECTION (工源三仟) (≈0.05‰; 161 / 3,400,000+ teams)

2023 **3rd Place** · CVPR 2023 PVUW Challenge, Video Semantic Segmentation Track

2022 **CVPR Outstanding Reviewer** (≈2%; 156 / 6,247)

2022 **Huawei Camera Academic Rising Star Award** (8 winners worldwide; 170,000+ live viewers; ¥30,000 bonus)

2022 **MICCAI Challenge: Glaucoma Oct Analysis and Layer Segmentation (GOALS)** · Team: IIAU-Segmentors · Glaucoma Detection 1st/100 · OCT Segmentation 3rd/100 · Overall 2nd/100 ($1,300)

2020 Huawei Fellowship

2018 **3rd Place · OPPO Top AI Competition** (Portrait Segmentation; 3 / 456; ¥50,000 bonus)

Publications Google Scholar: 5,000+ citations · ⋆: equal contribution · #: corresponding author

(Scopus ID: 57219626008. Public code available for all my first/co-first/last author works.)

TPAMI / IJCV / TIP	7
NeurIPS / ICML / CVPR / ECCV / ICCV / AAAI / MM	27 (3 oral, 1 highlight)
Papers with 1,000+ Citations	1
Papers with 500+ Citations	4
Papers with 100+ Citations	10
Papers as First / co-First / Last / Corresponding Author	25

Preprints

ConceptSeg-R1: Segment Any Concept via Meta-Reinforcement Learning

[44] Yuan Zhao, Youwei Pang, Jiaming Zuo, Wei Ji, Kailai Zhou, Bin Fan, Yunkang Cao, Lihe Zhang, Xiaofeng Liu, Huchuan Lu, Weisi Lin, Dacheng Tao, **Xiaoqi Zhao**#

arXiv 2026 [paper] [Code]

Seeing Through the Glare: A Multi-Source Benchmark and Ocular-Adaptive Pixel MeanFlow for Eyeglass Reflection Removal

[43] Tao Liu, Youwei Pang, Kailai Zhou, Jiaming Zuo, Hanqi Liu, Wei Ji, Peng-Tao Jiang, Xiaofeng Liu, Weisi Lin, **Xiaoqi Zhao**

arXiv 2026 [paper]

Hard, Yet Reducible: Controlled Forward Transfer for Synthetic Degradation Curation

[42] Chunming He, Kailai Zhou, Jiaming Zuo, Hanqi Liu, Fengyang Xiao, Youwei Pang, Xiaofeng Liu, Weisi Lin, **Xiaoqi Zhao**

arXiv 2026 [paper]

Satellite-Free Training for Drone-View Geo-Localization

[41] Tao Liu, Yingzhi Zhang, Xingrun Wang, Kan Ren, **Xiaoqi Zhao**

arXiv 2026 [paper]

Code-Interactive Visual Reasoning for Financial Candlestick Understanding

[40] Chenxi Zhang, Youwei Pang, **Xiaoqi Zhao**, Ziliang Gan, Rongjunchen Zhang, Shanshan Feng, Hao Wang

arXiv 2026 [paper]

Inspiring the Next Generation of Segment Anything Models: Comprehensively Evaluate SAM and SAM 2 with Diverse Prompts Towards Context-Dependent Concepts under Different Scenes

[39] **Xiaoqi Zhao**, Youwei Pang, Shijie Chang, Yuan Zhao, Lihe Zhang, Chenyang Yu, Hanqi Liu, Jiaming Zuo, Jinsong Ouyang, Weisi Lin, Georges El Fakhri, Huchuan Lu, Xiaofeng Liu

arXiv 2025 [paper] [Code]

Journals

Power Battery Detection

[38] **Xiaoqi Zhao**, Peiqian Cao, Chenyang Yu, Zonglei Feng, Lihe Zhang, Hanqi Liu, Jiaming Zuo, Youwei Pang, Jinsong Ouyang, Weisi Lin, Georges El Fakhri, Huchuan Lu, Xiaofeng Liu

IJCV 2026 [paper] [Code]

M^{2}SNet: Multi-scale in Multi-scale Subtraction Network for Medical Image Segmentation

[37] **Xiaoqi Zhao**, Hongpeng Jia, Youwei Pang, Long Lv, Feng Tian, Lihe Zhang, Weibing Sun, Huchuan Lu

MIR 2026 [paper] [Code] **2nd Place / 100 · MICCAI 2022 GOALS Challenge**

GOALS Challenge: Glaucoma OCT Analysis and Layer Segmentation

[36] Xiang Li, Wei Li, Huihui Fang, Chubin Ou, Xifei Wei, **Xiaoqi Zhao**, Youwei Pang, Dingyi Rong, Ziyu Zhou, Dechao Sun, Qiaobin Lin, Juntao Jiang, Qi Wei, Limin Huang, Zhihui Tian, Weili Jiang, Lin An, Yi Gou, Chengyun Song, Botond Fazekas, Taha Emre, Yihao Li, Yubo Tan, Yongjie Li, Du Lang, Jue Chen, Hufei Li, Xiaoyang Ye, Jun Wu, Lixin Duan, Huazhu Fu, Fei Li, Yanwu Xu, Xiulan Zhang

IEEE JBHI 2026 [paper]

DifferSeg: Towards Diverse Multimodal Binary Segmentation via Differential Perception and Frequency Guidance

[35] Qiangqiang Zhou, Jiawei Xu, Yong Chen, Dandan Zhu, Yugen Yi, **Xiaoqi Zhao**$^{\#}$

IEEE TCSVT 2026 [paper] [Code]

Classification and Calibration: Dual-Guidance Diffusion IModel for Mitigating Reconstruction Hallucinations in Multi-Class Anomaly Detection

[34] Yuan Zhao, **Xiaoqi Zhao**, Fan Yang, Lihe Zhang, Huchuan Lu

IEEE TCSVT 2026 [paper] [Code]

ComPtr: Towards Diverse Bi-source Dense Prediction Tasks via a General Complementary Transformer

[33] Youwei Pang, **Xiaoqi Zhao**, Lihe Zhang, Huchuan Lu

IEEE TPAMI 2025 [paper] [Code]

Beyond Mask: Rethinking Guidance Types in Few-shot Segmentation

[32] Shijie Chang, Youwei Pang, **Xiaoqi Zhao**, Huchuan Lu, Lihe Zhang

Pattern Recognition 2025 [paper] [Code]

Towards Diverse Binary Segmentation via A Simple yet General Gated Network

[31] **Xiaoqi Zhao**, Youwei Pang, Lihe Zhang, Huchuan Lu, Lei Zhang

IJCV 2024 [paper] [Code]

Adaptive Multi-source Predictor for Zero-shot Video Object Segmentation

[30] **Xiaoqi Zhao***, Shijie Chang*, Youwei Pang, Jiaxing Yang, Lihe Zhang, Huchuan Lu

IJCV 2024 [paper] [Code]

ZoomNeXt: A Unified Collaborative Pyramid Network for Camouflaged Object Detection

[29] Youwei Pang*, **Xiaoqi Zhao***, Tian-Zhu Xiang, Lihe Zhang, Huchuan Lu

IEEE TPAMI 2024 [paper] [Code]

CAVER: Cross-Modal View-Mixed Transformer for Bi-Modal Salient Object Detection

[28] Youwei Pang, **Xiaoqi Zhao**, Lihe Zhang, Huchuan Lu

IEEE TIP 2023 [paper] [Code] **ESI Highly Cited · Top 1%**

Joint Learning of Salient Object Detection, Depth Estimation and Contour Extraction

[27] **Xiaoqi Zhao**, Youwei Pang, Lihe Zhang, Huchuan Lu

IEEE TIP 2022 [paper] [Code]

Conferences

Localizing and Repairing Sparse-Prompt Failure in SAM Decoders via Box-to-Point Counterfactual

[26] Xiaobing Yu, Peijie Qiu, Jin Yang, Gordon Zhaoqi An, Weiwei Ma, Xiao Wu, Xuanzhao Dong, Wenhui Zhu, Xiwen Chen, **Xiaoqi Zhao**#, Xiaofeng Liu

NeurIPS 2026 (accepted)

PM-LoRA: Scalable Continual Learning via Progressive Merging of Low Rank Adapters

[25] Xiaobing Yu, Peijie Qiu, Jin Yang, **Xiaoqi Zhao**, Xuanzhao Dong, Wenhui Zhu, Xiwen Chen, Weiwei Ma, Xiaofeng Liu, Jiangpeng He

NeurIPS 2026 (accepted)

LadderEdit: Edit-Level Residual Compression for Memory-Efficient Lifelong Editing of LLMs

[24] Xiaobing Yu, Peijie Qiu, Jin Yang, Xuanzhao Dong, Weiwei Ma, Zhaoqi An, **Xiaoqi Zhao**#, Xiaofeng Liu

EMNLP 2026 [paper] [Code]

ActiveMedAgent: Cost-Aware Trajectory Learning for Multimodal Medical Diagnosis

[23] Weiwei Ma, Xiaobing Yu, Peijie Qiu, Jin Yang, Zhaoqi An, Xuanzhao Dong, **Xiaoqi Zhao**#, Xiaofeng Liu

EMNLP 2026 [paper]

UniMMAD: Unified Multi-Modal and Multi-Class Anomaly Detection via MoE-Driven Feature Decompression

[22] Yuan Zhao*, Youwei Pang*, Lihe Zhang#, Hanqi Liu, Jiaming Zuo*, Huchuan Lu, **Xiaoqi Zhao**#

CVPR 2026 [paper] [Code]

TP-Seg: Task-Prototype Framework for Unified Medical Lesion Segmentation

[21] Jiawei Xu, Qiangqiang Zhou#, Dandan Zhu, Yong Chen, Yugen Yi, **Xiaoqi Zhao**#

CVPR 2026 [Code]

SAM3-I: Segment Anything with Instructions

[20] Jingjing Li, Yue Feng, Yuchen Guo, Jincai Huang, Yongri Piao, Qi Bi, Miao Zhang, **Xiaoqi Zhao**, Qiang Chen, Shihao Zou, Wei Ji, Huchuan Lu, Li Cheng

ACL 2026 [paper] [Code]

U-Harmony: Enhancing Joint Training for Segmentation Models with Universal Harmonization

[19] Weiwei Ma, Xiaobing Yu, Peijie Qiu, Jin Yang, Pan Xiao, **Xiaoqi Zhao**, Xiaofeng Liu, Tomo Miyazaki, Shinichiro Omachi, Yongsong Huang

ISBI 2026 [paper] **Oral**

UniMRSeg: Unified Modality-Relax Segmentation via Hierarchical Self-Supervised Compensation

[18] **Xiaoqi Zhao**, Youwei Pang, Chenyang Yu, Lihe Zhang, Huchuan Lu, Shijian Lu, Georges El Fakhri, Xiaofeng Liu

NeurIPS 2025 [paper] [Code]

Rethinking Evaluation of Infrared Small Target Detection

[17] Youwei Pang, **Xiaoqi Zhao**#, Lihe Zhang, Huchuan Lu, Georges El Fakhri, Xiaofeng Liu, Shijian Lu

NeurIPS 2025 [Paper] [Code]

Unified Medical Lesion Segmentation via Self-referring Indicator

[16] Shijie Chang, **Xiaoqi Zhao**, Lihe Zhang, Tiancheng Wang

CVPR 2025 [Paper] [Code]

Mixture-of-Shape-Experts (MoSE): Shape Dictionary Framework to Prompt SAM for Medical Segmentation

[15] Jia Wei, **Xiaoqi Zhao**, Jonghye Woo, Jinsong Ouyang, Georges El Fakhri, Qingyu Chen, Xiaofeng Liu

CVPR Workshops 2025 [Paper]

Spider: A Unified Framework for Context-dependent Concept Segmentation

[14] **Xiaoqi Zhao**⋆, Youwei Pang⋆, Wei Ji⋆, Baicheng Sheng, Lihe Zhang, Jiaming Zuo, Huchuan Lu

ICML 2024 [paper] [Code]

Towards Automatic Power Battery Detection: New Challenge, Benchmark Dataset and Baseline

[13] **Xiaoqi Zhao**⋆, Youwei Pang⋆, Zhenyu Chen, Qian Yu, Lihe Zhang, Hanqi Liu, Jiaming Zuo, Huchuan Lu

CVPR 2024 [paper] [Code]

Multi-view Aggregation Network for Dichotomous Image Segmentation

[12] Qian Yu*, **Xiaoqi Zhao***, Youwei Pang*, Lihe Zhang, Huchuan Lu

CVPR 2024 [paper] [Code] **Highlight ≈3% [324 / 11,532]**

Open-Vocabulary Camouflaged Object Segmentation

[11] Youwei Pang*, **Xiaoqi Zhao***, Jiaming Zuo, Lihe Zhang, Huchuan Lu

ECCV 2024 [paper] [Code]

Isomer: Isomerous Transformer for Zero-shot Video Object Segmentation

[10] Yichen Yuan, Yifan Wang, Lijun Wang, **Xiaoqi Zhao**, Huchuan Lu, et al.

ICCV 2023 [paper] [Code]

Zoom In and Out: A Mixed-scale Triplet Network for Camouflaged Object Detection

[9] Youwei Pang*, **Xiaoqi Zhao***, Tian-zhu Xiang, Lihe Zhang, Huchuan Lu

CVPR 2022 [paper] [Code]

Self-Supervised Pretraining for RGB-D Salient Object Detection

[8] **Xiaoqi Zhao**, Youwei Pang, Lihe Zhang, Huchuan Lu, Xiang Ruan

AAAI 2022 [paper] [Code] **Poster ≈15% [1,349 / 9,020]**

Multi-Source Fusion and Automatic Predictor Selection for Zero-Shot Video Object Segmentation

[7] **Xiaoqi Zhao**, Youwei Pang, Jiaxing Yang, Lihe Zhang, Huchuan Lu

ACM MM 2021 [paper] [Code] **Oral ≈9%**

Automatic Polyp Segmentation via Multi-scale Subtraction Network

[6] **Xiaoqi Zhao**, Lihe Zhang, Huchuan Lu

MICCAI 2021 [paper] [Code] **Top 1% Cited · Ranking #2 in Colonoscopy @ MICCAI 2021**

Suppress and Balance: A Simple Gated Network for Salient Object Detection

[5] **Xiaoqi Zhao***, Youwei Pang*, Lihe Zhang, Huchuan Lu, Lei Zhang

ECCV 2020 [paper] [Code] **Oral ≈2%** **Top 3% Cited · Ranking #1 in Visual Saliency @ ECCV 2020**

A Single Stream Network for Robust and Real-time RGB-D Salient Object Detection

[4] **Xiaoqi Zhao**, Youwei Pang, Lihe Zhang, Huchuan Lu, Lei Zhang

ECCV 2020 [paper] [Code]

Multi-scale Interactive Network for Salient Object Detection

[3] Youwei Pang*, **Xiaoqi Zhao***, Lihe Zhang, Huchuan Lu

CVPR 2020 [paper] [Code] **Top 4% Cited · Ranking #1 in Visual Saliency @ CVPR 2020**

Hierarchical Dynamic Filtering Network for RGB-D Salient Object Detection

[2] Youwei Pang, Lihe Zhang, **Xiaoqi Zhao**, Huchuan Lu

ECCV 2020 [paper] [Code]

3rd Place Solution for PVUW2023 VSS Track: A Large Model for Semantic Segmentation on VSPW

[1] Shijie Chang, Zeqi Hao, Ben Kang, **Xiaoqi Zhao**, et al., Huchuan Lu

[paper] [Code] **3rd Place · CVPR 2023 PVUW Challenge**

Well-recognized Open-source Projects / Toolkits / Awesomes

GitHub Followers: 400+ · Total Stars: 4,000+

(All my accepted papers as First/co-First/Last author have their code released on GitHub.)

Awesome-AI4X-NSCLR-Papers ★136

AI+X papers in Nature / Science / Cell / Lancet / Radiology

Awesome-Open-Vision-Problem-Modeling ★140

Awesome List for Diverse Open Vision Problems and Modeling Solutions

Awesome-Unified-Context-dependent-Concept-Segmentation ★119

Awesome List for Unified Context-dependent Concept Segmentation

SAMs-CDConcepts-Eval ★575

Comprehensive Evaluation of SAM & SAM 2 with Diverse Prompts

ConceptSeg-R1 ★244

Segment Any Concept via Meta-Reinforcement Learning

Spider-UniCDSeg ★355

(ICML 2024) Unified Framework for Context-dependent Concept Segmentation

MSNet-M2SNet ★1,030

(MICCAI 2021 & 2022 GOALS) Automatic Polyp Segmentation via Multi-scale Subtraction Network

X-ray-PBD ★319

(IJCV 2026 & CVPR 2024) Power Battery Detection Benchmark

GateNet-RGB-Saliency ★263

(IJCV 2024 & ECCV 2020 Oral) Diverse Binary Segmentation

MS-APS ★121

(IJCV 2024 & MM 2021 Oral) Multi-Source Fusion and Automatic Predictor Selection for Zero-Shot Video Object Segmentation

UniMRSeg ★180

(NeurIPS 2025) Unified Modality-Relax Segmentation

MMFT ★111

(IEEE TIP 2022) Joint Learning of Salient Object Detection, Depth Estimation and Contour Extraction

SSLSOD ★103

(AAAI 2022) Self-Supervised Pretraining for RGB-D Salient Object Detection

DANet ★109

(ECCV 2020) A Single Stream Network for Robust and Real-time RGB-D Salient Object Detection

Participated Grants / Projects

Singapore National Research Foundation (NRF) Postdoctoral Award Grant Mar. 2026 – Mar. 2030

Principal Investigator (PI) · SGD $250,000

Singapore's most prestigious postdoctoral funding, awarded by the National Research Foundation under the Prime Minister's Office. Open to applicants worldwide; the fellowship supports only a handful of outstanding researchers each year and is explicitly structured as a pathway toward an independent faculty/research position in Singapore. Comes with full autonomy over research direction and lab establishment budget.

NTU AI-for-X (AI4X) Postdoctoral Fellowship Grant Mar. 2026 – Mar. 2030

Principal Investigator (PI) · Nanyang Technological University

Awarded by NTU, supporting the AI4X initiative to advance general-purpose AI across scientific and engineering disciplines.

X3000Former Series — Industrial Vision Foundation Models Industry Project 2021.06 – 2024.12

Chief Scientist · X3000 INSPECTION (Startup, Series B Funded) · Clients: CATL, Ganfeng Lithium, EVE Energy

- **Unified Industrial Imaging & Inspection System for Top Power Battery Companies:** Fast High-resolution CT Reconstruction · Unified Visual-Language Model · Fine-grained Image Generation
- **Power Battery Inspection & Enhancement for Top Power Battery Companies:** Tiny Object Detection · Overhang Measurement · Image Blind Enhancement
- **Automation of Production Systems for Top Automobile OEMs:** OCR · Internal Defect Detection

Professional Services

Conference Area Chair / Reviewer

- NeurIPS 2026 **Area Chair**
- ACL 2026
- CVPR 2022–2026 **(Outstanding Reviewer ≈2%)**
- ICCV 2021–2025
- ECCV 2022–2024
- AAAI 2022–2026
- NeurIPS 2023–2025
- ICLR 2024–2026
- ICML 2024–2026

Journal Reviewer

- IEEE TPAMI
- International Journal of Computer Vision (IJCV)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Medical Imaging (TMI)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Multimedia (TMM)